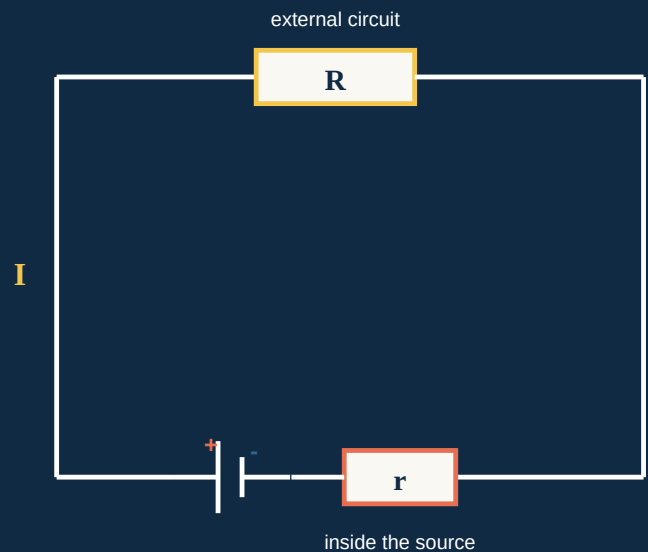


Ohm's Law for Closed Circuits

A reasoning-first guide to real sources, internal resistance, terminal voltage, and power.

A real source does two things at once:
it supplies electrical energy and
consumes some of that energy
internally.



$$E = I(R + r)$$

BY THE END OF THIS HANDOUT, YOU SHOULD BE ABLE TO

- explain EMF as energy supplied per unit charge;
- derive and apply the closed-circuit form of Ohm's law;
- interpret terminal voltage, U-I graphs, power, and efficiency;
- determine EMF and internal resistance from measurements.

Why does a battery become warm?

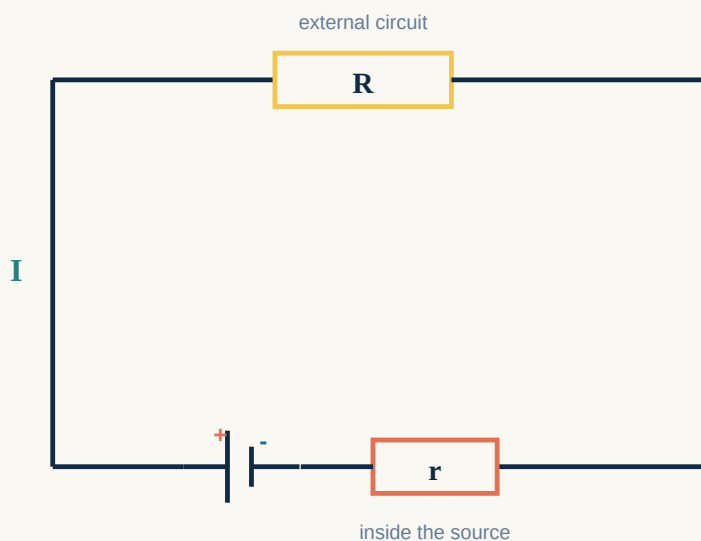
The heating is evidence that a real source has resistance inside it.



OBSERVATION

A battery that supplies a large current often becomes warm, even when the external wires and load are working normally.

The simplest useful model



INTERNAL HEATING

$$Q_{\text{internal}} = I^2 r t$$

The internal resistance r converts electrical energy into thermal energy while current flows.

KEY MODELING DECISION

We treat the source as an ideal energy supplier with EMF E , connected in series with an internal resistance r . The rest of the circuit is represented by the external resistance R .

EMF is energy supplied per unit charge.

It describes the source's ability to convert other forms of energy into electrical energy.

$$E = W_{\text{non-electrostatic}} / q$$

1 volt = 1 joule per coulomb

EMF is measured in volts, but it is not simply the voltage across a working source.

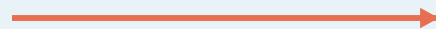
What provides the non-electrostatic force?



ENERGY CONVERSION

In a chemical cell, chemical forces move charge inside the source. In a generator, mechanical forces do the same job.

chemical or mechanical energy



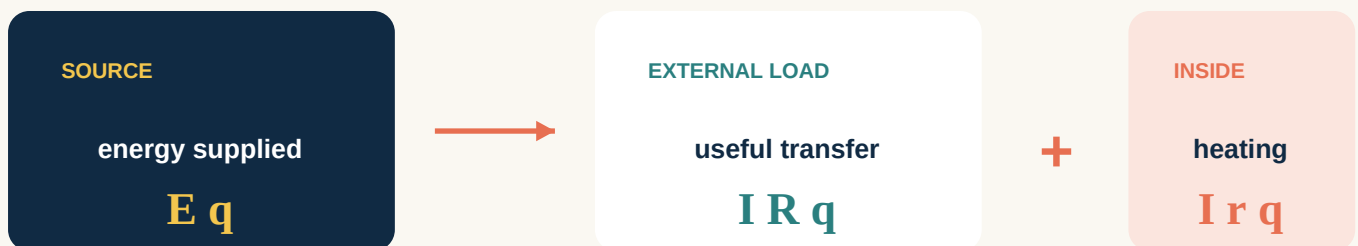
electrical energy

DO NOT CONFUSE THESE IDEAS

A larger EMF means more energy is supplied per coulomb. It does not necessarily mean that the source stores more total energy or delivers energy faster.

One energy balance produces the closed-circuit equation.

Follow one unit charge around the complete circuit.



$$E q = I R q + I r q$$

$$E = I(R + r)$$

Closed-circuit Ohm's law

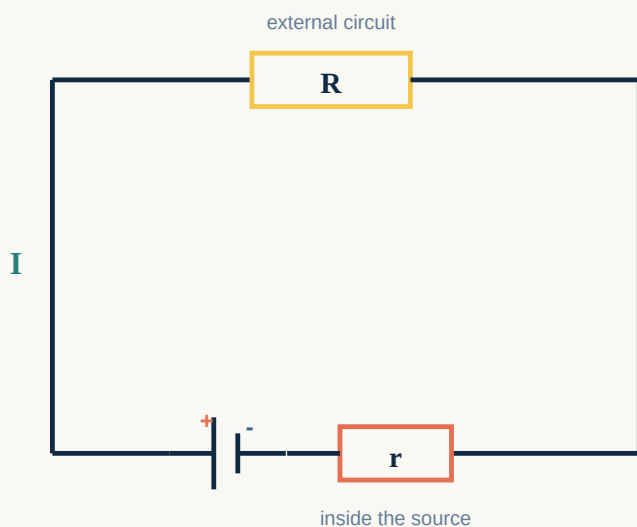
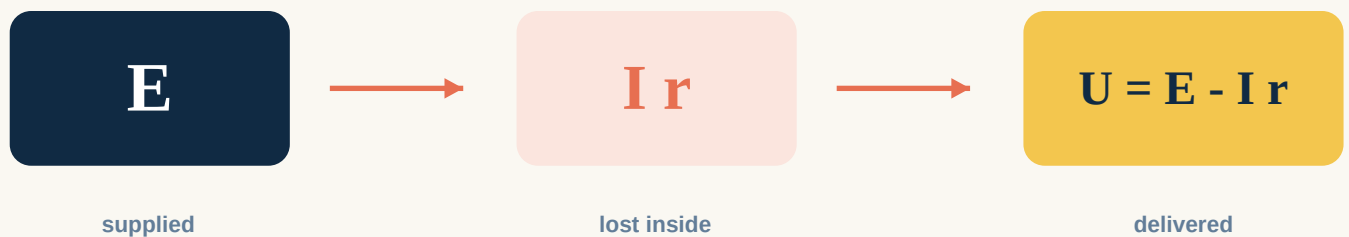
$$I = E / (R + r)$$

INTERPRETATION

The current is controlled by the total series resistance. Increasing either the load resistance R or the internal resistance r reduces the current.

The load receives what remains after the internal voltage drop.

Terminal voltage is the voltage available to the external circuit.



EQUIVALENT EXPRESSIONS

$$U = E - I r$$

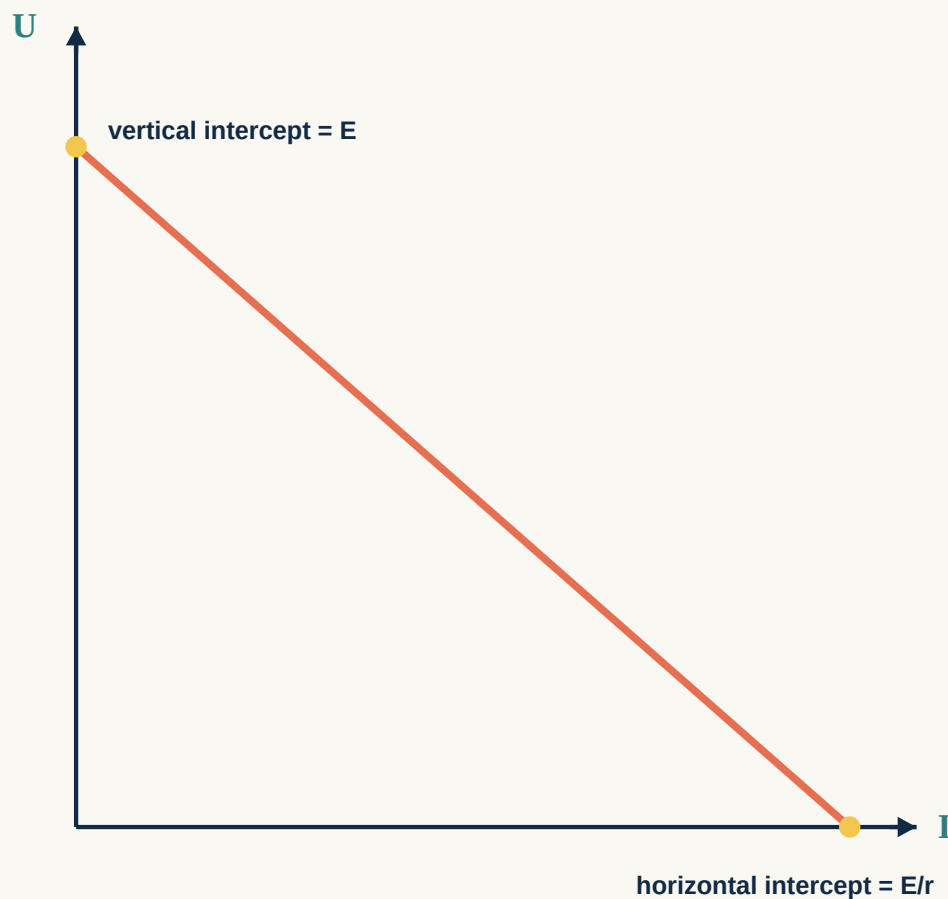
$$U = I R$$

THREE IMPORTANT CASES

- Open circuit: $I = 0$, so $U = E$.
- Normal operation: U is less than E because $I r > 0$.
- Short circuit: R is nearly zero, so current can become dangerously large.

The U-I graph reveals both E and r.

The equation $U = E - I r$ has the form of a straight line.



SLOPE

$$\Delta U / \Delta I = -r$$

HOW TO READ IT

A steeper downward line means a larger internal resistance.

EXPERIMENTAL USE

Measure several pairs of current and terminal voltage. Fit a straight line: the intercept gives E , while the magnitude of the slope gives r .

The source power splits into useful output and internal heating.

Power accounting makes the role of internal resistance visible.

$$P_{\text{source}} = E I$$

=

$$P_{\text{load}} = I^2 R$$

+

$$P_{\text{internal}} = I^2 r$$

Efficiency

$$\eta = P_{\text{load}} / P_{\text{source}} = R / (R + r)$$

HIGH EFFICIENCY

When R is much larger than r , most of the source energy reaches the load.

HEAVY CURRENT DEMAND

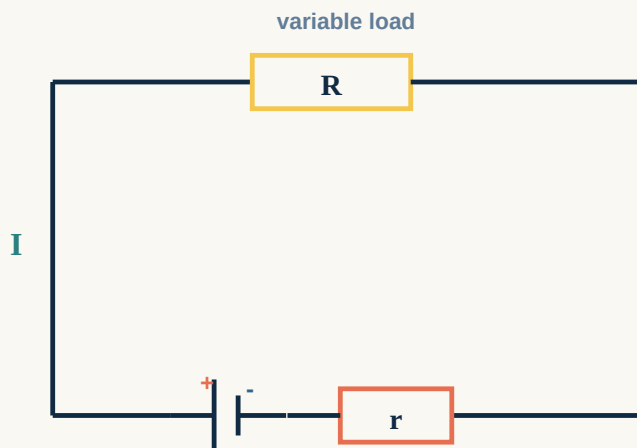
When R is small, current is large and internal heating can become significant.

EXTENSION: MAXIMUM POWER TRANSFER

The load receives maximum power when $R = r$. At that point, however, only half of the source power reaches the load, so maximum power is not maximum efficiency.

Two operating points are enough to determine E and r .

Change the load, then measure current and terminal voltage.



MEASUREMENTS

$$U_1 = E - I_1 r$$

$$U_2 = E - I_2 r$$

$$r = (U_1 - U_2) / (I_2 - I_1)$$

$$E = U_1 + I_1 r$$

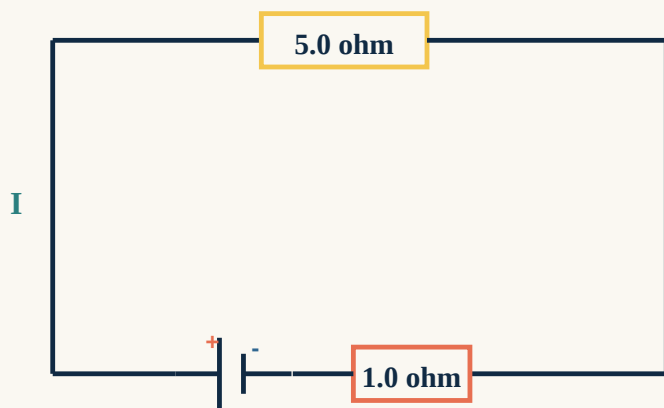
A practical shortcut

OPEN-CIRCUIT MEASUREMENT

With an ideal voltmeter and no current, $U_{\text{open}} = E$. Then connect a known load, measure I and U , and calculate $r = (E - U)/I$.

A single model answers current, voltage, power, and efficiency.

A source has $E = 12\text{ V}$ and $r = 1.0\text{ ohm}$. It is connected to a 5.0 ohm load.



1 · Current

$$I = E/(R + r) = 12/(5.0 + 1.0) = 2.0\text{ A}$$



2 · Terminal voltage

$$U = I R = 2.0 \times 5.0 = 10\text{ V}$$



3 · Internal heating

$$P_{\text{internal}} = I^2 r = 4.0\text{ W}$$



4 · Useful load power

$$P_{\text{load}} = I^2 R = 20\text{ W}$$



5 · Efficiency

$$\eta = R/(R + r) = 5/6 = 83\%$$

CHECK

Energy accounting is consistent: $P_{\text{source}} = E I = 24\text{ W}$, while $P_{\text{load}} + P_{\text{internal}} = 20\text{ W} + 4\text{ W} = 24\text{ W}$.

Measurements reveal an aging battery's internal resistance.

The open-circuit voltage is 1.50 V. With a load connected, $U = 1.20$ V and $I = 0.60$ A.

$$E = U_{\text{open}} = 1.50 \text{ V}$$

$$r = (E - U) / I$$

$$r = (1.50 - 1.20) / 0.60 = 0.50 \text{ ohm}$$

PHYSICAL MEANING

The battery loses 0.30 V internally while supplying 0.60 A. A larger internal resistance causes a larger voltage drop under load and is a common sign of battery aging.

DIAGNOSTIC RULE

A source may show a normal open-circuit voltage but still perform poorly under load. Always test both voltage and current behavior.

Use the model before reaching for a formula.

For each question, identify the source, the internal resistance, and the external circuit.

- 1 A 6.0 V source with $r = 1.0 \text{ ohm}$ is connected to $R = 2.0 \text{ ohm}$. Find I , U , P_{internal} , and efficiency.
- 2 A source has $E = 3.0 \text{ V}$. When it supplies 0.40 A , its terminal voltage is 2.6 V . Find r .
- 3 A U - I graph crosses the U -axis at 9.0 V and the I -axis at 3.0 A . Determine E and r .
- 4 Explain why the terminal voltage approaches the EMF when the external resistance becomes very large.
- 5 Two sources have the same EMF, but source A has a larger internal resistance. Compare their terminal voltages when they supply the same current.
- 6 A load is changed so that R becomes equal to r . What fraction of the source power reaches the load? Is this maximum efficiency?

Hint: begin with $E = I(R + r)$, $U = E - I r$, and $\eta = R/(R + r)$.

The complete model in one page.

A real source is an energy converter with an unavoidable internal resistance.

CORE MODEL

ideal EMF E in series with internal resistance r

CURRENT

$$I = E / (R + r)$$

TERMINAL VOLTAGE

$$U = E - I r = I R$$

POWER SPLIT

$$E I = I^2 R + I^2 r$$

EFFICIENCY

$$\eta = R / (R + r)$$

PRACTICE ANSWERS

1. $I = 2.0 \text{ A}$; $U = 4.0 \text{ V}$; $P_{\text{internal}} = 4.0 \text{ W}$; $\eta = 67\%$.
2. $r = 1.0 \text{ ohm}$.
3. $E = 9.0 \text{ V}$; $r = 3.0 \text{ ohm}$.
4. As R grows, I approaches zero, so $I r$ approaches zero and U approaches E .
5. Source A has the lower terminal voltage because its internal drop $I r$ is larger.
6. One half reaches the load. This is maximum load power, not maximum efficiency.

Central question: where does each joule per coulomb go?